20 The equation of a curve is $y = a^x$
A is the point where the curve intersects the y-axis.

(a) State the coordinates of A.

On the y-axis, x = 0 and $a^0 = 1$.

1 mark
The y-intercept is (0, 1)

(0 , 1)

(1)

The equation of circle **C** is $x^2 + y^2 = 16$

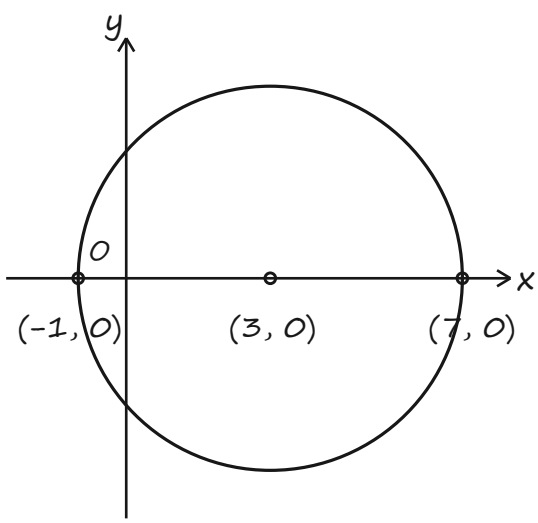
The circle **C** is translated by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ to give circle **B**.

(b) Draw a sketch of circle **B**.

Label with coordinates
the centre of circle **B**
and any points of intersection with the x-axis.

Circle C: centre (0, 0), radius √16 = 4.

Translate 3 units right: centre (3, 0), radius still 4.



1 mark
Move the centre to (3, 0)

1 mark
Keep radius 4 and find intercepts

1 mark
Correct circle and all three labels

x-intercepts: 3 − 4 = −1 and 3 + 4 = 7.

(3)

(Total for Question 20 is 4 marks)